DYP DPU
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**Department of Computer Engineering**

**Subject Name: Artificial Intelligence**

# Practical No. 1

**Title:**

**Design and Implement a Simple Reflex Agent for the Vacuum Cleaner World and Analyze Its Rational Behavior in a Given Environment**

**Aim**

To design and implement a **Simple Reflex Agent** for the Vacuum Cleaner World and analyze its rational behavior based on the current percept of the environment.

**Objectives**

1. To understand the concept of an **intelligent agent**.
2. To understand the working of a **Simple Reflex Agent**.
3. To implement a Vacuum Cleaner Agent using **Python**.
4. To analyze the agent's actions for different environmental conditions.
5. To understand the concept of **rational behavior** of an AI agent.

**Theory:**

**What is an Intelligent Agent?**

An intelligent agent is a system that perceives its environment through sensors and performs actions through actuators to achieve a goal.

**Basic structure:**

**Environment → Sensors → Agent → Actuators → Environment**

For a vacuum cleaner:

- **Sensors:** Detect the current location and whether the floor is clean or dirty.
- **Agent:** Decides what action to perform.

- **Actuators:** Move the vacuum cleaner or perform cleaning.
- **Environment:** Rooms/floor locations containing clean or dirty surfaces.

**Vacuum Cleaner World**

Consider a simple environment containing two locations:

- Location A
- Location B

Each location can have one of two states:

- Clean
- Dirty

The vacuum cleaner can perform the following actions:

| Condition | Action |
|---|---|
| Current location is Dirty | Suck |
| Current location is Clean and agent is at A | Move Right |
| Current location is Clean and agent is at B | Move Left |

**Example**

Suppose the initial environment is:

| Location | Status |
|---|---|
| A | Dirty |
| B | Dirty |

Agent starts at **A**.

The agent performs:

**A Dirty → Suck → A Clean → Move Right → B Dirty → Suck → B Clean**

Thus, the agent cleans both locations.

## Simple Reflex Agent

A Simple Reflex Agent selects an action based only on the current percept.

It uses condition-action rules:

**IF condition THEN action**

For example:

- IF current location is dirty → **Suck**
- IF current location is clean and agent is at A → **Move Right**
- IF current location is clean and agent is at B → **Move Left**

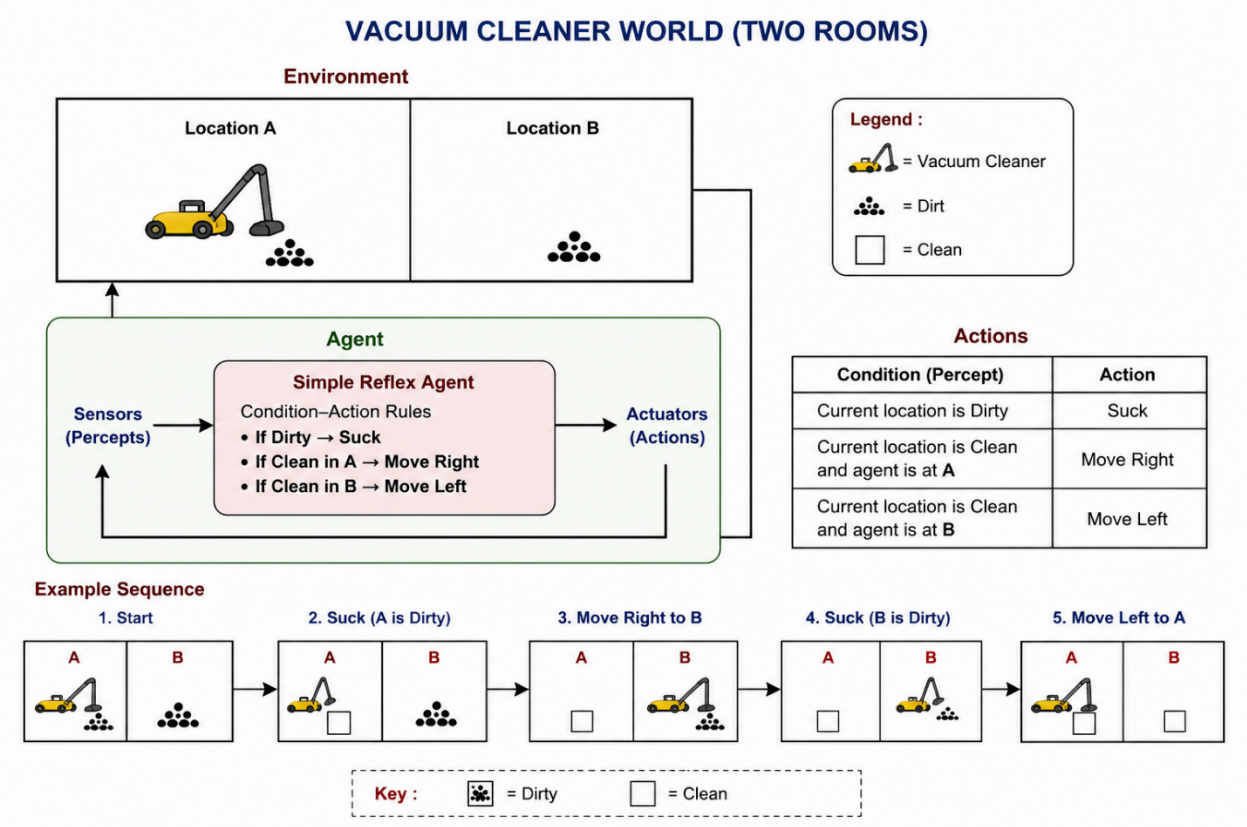
The agent does not maintain a detailed history of previous states.

## PEAS Description

PEAS stands for:

P – Performance Measure
E – Environment
A – Actuators
S – Sensors

| Component | Vacuum Cleaner Agent |
|---|---|
| **Performance Measure** | Maximum cleanliness, minimum unnecessary movement, minimum time/energy |
| **Environment** | Two-room environment: A and B |
| **Actuators** | Suck, Move Left, Move Right |
| **Sensors** | Current location and cleanliness status |




## Algorithm

**Algorithm: Simple Reflex Vacuum Cleaner Agent**

**Step 1:** Start.

**Step 2:** Define the environment with two locations A and B.

**Step 3:** Set the initial position of the vacuum cleaner.

**Step 4:** Read the current location and its cleanliness status.

**Step 5:** If the current location is dirty, perform **Suck**.

**Step 6:** If the current location is clean:

- If the agent is at A, move to B.
- If the agent is at B, move to A.

**Step 7:** Update the environment after every action.

**Step 8:** Repeat the process for a specified number of steps.

**Step 9:** Display the actions performed and final environment.

**Step 10:** Stop.

**Program-**

**Output-**

**Conclusion-**

**13. Viva Questions**

1. What is an intelligent agent?
2. What is a Simple Reflex Agent?
3. What is the difference between an agent and an environment?
4. What are sensors and actuators?
5. What is PEAS?
6. What is the performance measure for a vacuum cleaner agent?
7. Why is the vacuum cleaner considered an intelligent agent?
8. What is meant by rational behavior?
9. What are the limitations of a Simple Reflex Agent?
10. How is a Model-Based Reflex Agent different from a Simple Reflex Agent?
11. Can a Simple Reflex Agent remember previous actions?
12. What happens if both rooms are clean?
13. Why is if-elif-else used in this implementation?
14. What are the possible percepts in the Vacuum Cleaner World?